Project Stage 1 (Midterm) — Concept, Requirements and Architecture of an ICT Project

Team: Salima Zimbekkyzy, Azhar Saparzholova, Nurassyl Oralbekov

Topic: Improvement of MyDU & LMS

Group: SE-2623

Part A – Problem definition and team charter

A1. Problem statement

As students who use MyDU, LMS, Learn, Microsoft Teams on a daily basis, we have noticed that information related to studying is distributed across several separate platforms. Students have to switch between different systems to check assignments, deadlines, attendance, schedules, communication, and other academic information. This fragmentation can make it difficult to find important information quickly and increases the chance of missing deadlines or notifications.

To investigate whether this problem affects other students, we conducted a survey among **20 first-year students**. Participants rated the convenience of having LMS, Learn, and MyDU as separate platforms on a scale from 1 to 10. The average score was only **2.95/10**, providing evidence that the current separation of platforms is inconvenient for many students. Several respondents also suggested placing deadlines and information from different platforms in one location, improving notifications, and integrating Lighthouse into MyDU.

Currently, students solve the problem by manually checking multiple platforms and remembering where specific information is located. ICT could solve this more effectively by integrating these services into a single platform with a unified dashboard, notifications, deadlines, and access to the existing systems. This would reduce unnecessary navigation and make academic information easier to access and manage.

Отметка времени	What do you use myDU for?	What do you use LMS for?	How convenient are separated platforms for you? (lms , learn and MyDU as separate platforms)	If you were responsible for platform development, what would you do?
2026/10/04 4:44:03 PM GMT+5	view homework deadlines;peer advisor	view assignments;check attendance	10	The AI bot could be faster, it often takes too long to get an answer.
2026/10/04 4:46:00 PM GMT+5	AI chatbot;peer advisor;view homework deadlines	check attendance;view the syllabus	5	A Mark as Read button for notifications would be really useful.
2026/10/04 4:50:02 PM GMT+5	view the schedule;AI chatbot;peer advisor	view assignments;view deadlines	1	It would be great to have all deadlines in one place. Another thing I'd improve is that it would be useful to see schedule changes immediately when a class is moved or cancelled. Finally, it would be convenient to move lighthouse functionality into mydu.
2026/10/04 4:56:15 PM GMT+5	view the schedule;peer advisor;AI chatbot	view the syllabus	3	I would move Lighthouse into MyDU.
2026/10/04 4:57:19 PM GMT+5	AI chatbot;view the schedule	view deadlines;view the syllabus;check attendance	3	The notification system could be organized better and the design could be more modern and intuitive.
2026/10/04 4:58:36 PM GMT+5	AI chatbot;view the schedule	check attendance;view deadlines;view assignments	6	It would be really useful to have a university map with geolocation, especially for finding classrooms. Also, notifications about upcoming deadlines and classes would be useful.
2026/10/04 5:03:50 PM GMT+5	peer advisor;view homework deadlines;view the schedule	view deadlines	4	The notification system shows that there are some unread notifications, even though there are no new notifications.
2026/10/04 5:05:18 PM GMT+5	view homework deadlines	view deadlines;view the syllabus;view assignments	2	The AI bot is too slow. Also, I'd change notification system and fix some small bugs
2026/10/04 5:09:05 PM GMT+5	view homework deadlines;view the schedule	view assignments;view the syllabus;check attendance	8	It would be nice to have information about available study spaces.
2026/10/04 5:14:43 PM GMT+5	view the schedule;peer advisor	view assignments	1	The academic calendar could be easier to navigate and understand. Moreover, there are some bugs that should be fixed. Finally, a search function for classrooms, teachers and university information would be helpful.
2026/10/04 5:15:52 PM GMT+5	AI chatbot;view the schedule	view the syllabus;view deadlines	3	it is tiring to scroll down to find dashboard in LMS on the phone. It would be great if we had opportunity to change pfp in mydu
2026/10/04 5:20:51 PM GMT+5	peer advisor;AI chatbot	view the syllabus;view deadlines	2	It would be convenient if Lighthouse was in MyDU.
2026/10/04 5:23:20 PM GMT+5	AI chatbot;view homework deadlines;view the schedule	check attendance	1	The mobile design of LMS could be more intuitive, and ai bot in mydu faster.
2026/10/04 5:24:19 PM GMT+5	peer advisor;view the schedule;view homework deadlines	check attendance;view the syllabus	1	It would be great to have all deadlines from all platforms in one place.
2026/10/04 5:25:43 PM GMT+5	view the schedule;peer advisor	check attendance;view assignments;view the syllabus	2	The AI bot is not useful. Apart from that, It would be nice if in LMS had a dark-theme, and mobile lms is inconvenient - every time very long scrolling down to find dashboard and my courses
2026/10/04 5:30:05 PM GMT+5	view homework deadlines;peer advisor;view the schedule	view assignments	1	navigation board on mobile version of LMS located at the very end
2026/10/04 5:34:19 PM GMT+5	view homework deadlines;peer advisor;view the schedule	view deadlines	1	sessions in LMS expire very fast. Quizzes from Learn could be integrated in LMS, I always forget about them
2026/10/04 5:35:34 PM GMT+5	view the schedule;view homework deadlines;peer advisor	view assignments;view the syllabus;check attendance	2	It would be really useful to have a university map with geolocation, especially for finding classrooms and free spaces
2026/10/04 5:38:17 PM GMT+5	AI chatbot;view homework deadlines;view the schedule	view deadlines;view the syllabus;check attendance	2	The notification system is very annoying
2026/10/04 5:39:43 PM GMT+5	view the schedule;peer advisor;AI chatbot	view assignments;view deadlines;view the syllabus	1	A Mark as Read button for notifications would be really useful. Another thing - better organization of notifications. Finally, i would like to see which classrooms are currently free.

A2. Users and scope

Our system has two group of users:

- **Students** - comfortable usage of applications and web-sites
- **Administrator** - want to achieve comfort for the students

Will Include	Will NOT Include
Integration of MyDU, LMS, Teams, Learn	Leave du.astanait.edu.kz as a separete platform
General improvements	Data imported to LMS and myDU
Fix bugs	-
Improve UX/UI	-

A3. Goals and success criteria

Goal of our project - achieve the comfortable usage of all platforms

3 success criteria:

- amount of complaints reduces by 45%,
- crash-free session rate sharply rises,
- number of critical bugs drops to zero

A4. Team Charter

Team Members	Role	Parts
Salima	Designer	A,D,E
Azhar	Analyst	B,E
Nurassyl	Developer	C,E

Part B — Requirements

ID	Requirement	Type	Priority	How it will be checked
FR1	**Single Sign-On (SSO):** The user logs in once using Microsoft university credentials to access MyDU, LMS, and Learn simultaneously.	Functional	Must	Test logging in with a student Microsoft account; verify seamless session establishment across all services.
FR2	**Direct Learn Quiz Integration:** Quizzes and assignment deadlines from Learn are imported and embedded directly into the LMS course interface.	Functional	Must	Open a course module in LMS; verify that Learn quizzes appear and can be completed directly within LMS without external redirects.
FR3	**Integrated Lighthouse Campus Navigation:** The Lighthouse module is integrated into MyDU, providing an interactive university map, classroom search, and real-time free room availability.	Functional	Must	Access Lighthouse via MyDU; search for a classroom and check if current free/occupied status and floor map render correctly.
FR4	**Centralized Notification System:** System delivers organized notifications with a manual "Mark all as read" button and persistent unread badge state.	Functional	Should	Trigger a notification; test clicking "Mark as Read" and verify unread badge count updates correctly.
FR5	**AITUSA Hub** (Events, Hackathons & Discounts): A dedicated section aggregating major student council events, hackathons, and exclusive student discounts provided by AITUSA partners.	Functional	Should	Open the AITUSA tab; verify event listings, hackathon registration links, and student discount details are visible.
FR6	**Student Clubs Directory:** A categorized directory of all university clubs with active links to their social media handles (Instagram, Telegram).	Functional	Should	Select a club from the directory; test clicking Instagram/Telegram links to ensure correct redirection.
FR7	**Integrated AI Chatbot:** Fast-responding AI assistant in MyDU answers student queries regarding schedules, teachers, and university navigation.	Functional	Could	Submit 5 queries to the chatbot; verify response generation time and accuracy.
FR8	**Redesigned Academic Calendar:** A	Functional	Could	Open the academic

	grid/card-based visual redesign of the academic calendar replacing dense text lists for clearer navigation.			calendar tab; verify event dates and exam periods display in a visually structured layout.
NFR1	**Performance:** Pages load and parse LMS/Learn data in under 3 seconds per request for up to 1,000 concurrent student users.	Non-functional	Must	Execute simulated load testing (e.g., JMeter/Locust) against FastAPI endpoints.
NFR2	**Security & Compliance:** All traffic is encrypted via HTTPS/TLS, with backend processing hosted in Kazakhstan (ps.kz) without storing passwords or persistent user data.	Non-functional	Must	Perform security audit to confirm no student credentials or grades are written to disk; verify Kazakhstani IP hosting.
NFR3	**Usability:** A mobile-first responsive layout with top navigation bar that eliminates excessive scrolling and retains login session for 8 hours.	Non-functional	Must	Test navigation ergonomics and "Remember Me" session persistence on iOS/Android browsers.
NFR4	**Availability:** System maintains 99.5% uptime per month, specifically guarantees 99.9% availability during peak exam and deadline periods.	Non-functional	Must	Monitor server uptime using UptimeRobot during high-traffic assignment submission windows.

Among the specified requirements, **FR2 (Direct Learn Quiz Integration)**, **NFR3 (Usability: Mobile-First Responsive Interface)**, and **NFR2 (Security & Kazakhstani Legal Compliance)** are the most critical. **FR2** directly resolves platform fragmentation by moving isolated Learn quizzes and assignment deadlines directly into the core LMS workflow so students no longer miss tasks. **NFR3** is essential because 75% of surveyed students complained about LMS's clumsy mobile layout and exhausting scrolling. Finally, **NFR2** ensures mandatory compliance with Kazakhstani data privacy laws by processing sessions locally on ps.kz servers without persistent credential storage. Together, these three requirements form the foundational pillar for user satisfaction, technical feasibility, and legal deployment.

Part C — Proposed solution and architecture

C1. Architecture diagram

As shown in Figure 1, the process starts on the client device. First, the client sends a request (to get the data using the user's Microsoft university account) over the network using HTTPS / WebSockets. The network redirects the request to the server / cloud side. The server can be our local device or a cloud server, for example from ps.kz (we can use ONLY servers located in Kazakhstan because of the law). On this step a headless machine logs in to the LMS with the user's login, using the official LMS API if it is provided, or Python Playwright / BeautifulSoup to access the data otherwise. After parsing, the data is sent on the fly as an HTTPS response to the user. When the user receives all the data fetched from his account, the browser (client side) renders it with a more comfortable UI and UX for mobile.

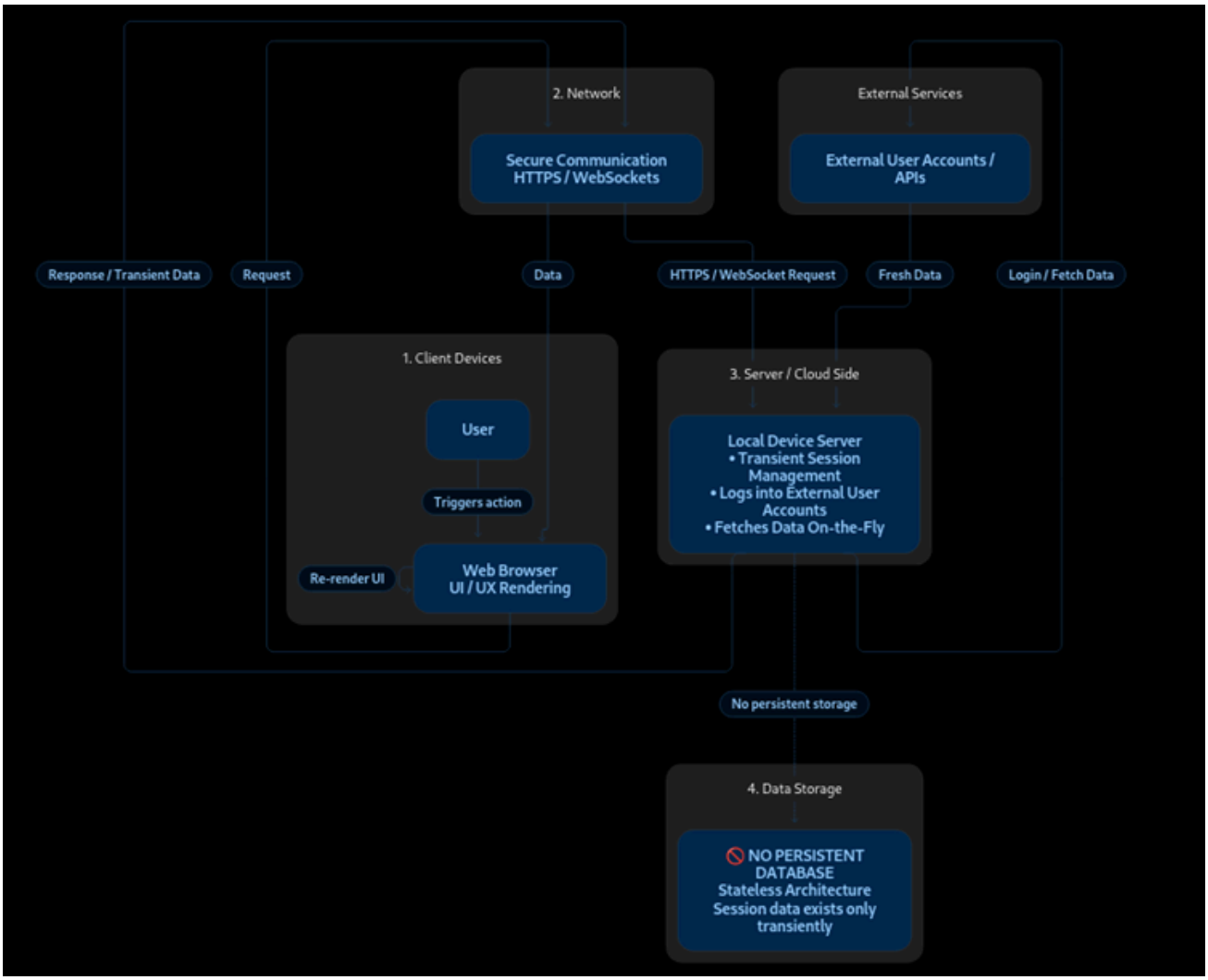


Figure 1. Architecture of the solution

Architectural style: client-server + cloud-based.

We chose the client-server style with a cloud-based server side. Cost: the interface is deployed for free on GitHub Pages, and only one small server in Kazakhstan is rented. Scalability: the server is stateless, so more instances can be added behind Nginx when the number of students grows. Performance: the browser only renders the data, while Python fetches it on the fly; a server inside Kazakhstan keeps the delay low. Security: Kazakhstan law does not allow taking citizens' personal data out of the country, so AWS and other foreign services are excluded; there is no persistent database, so credentials and grades are never stored and exist only in session memory. Maintenance: one web client works on every phone, and one Python code base is easy to update.

C2. Technology stack

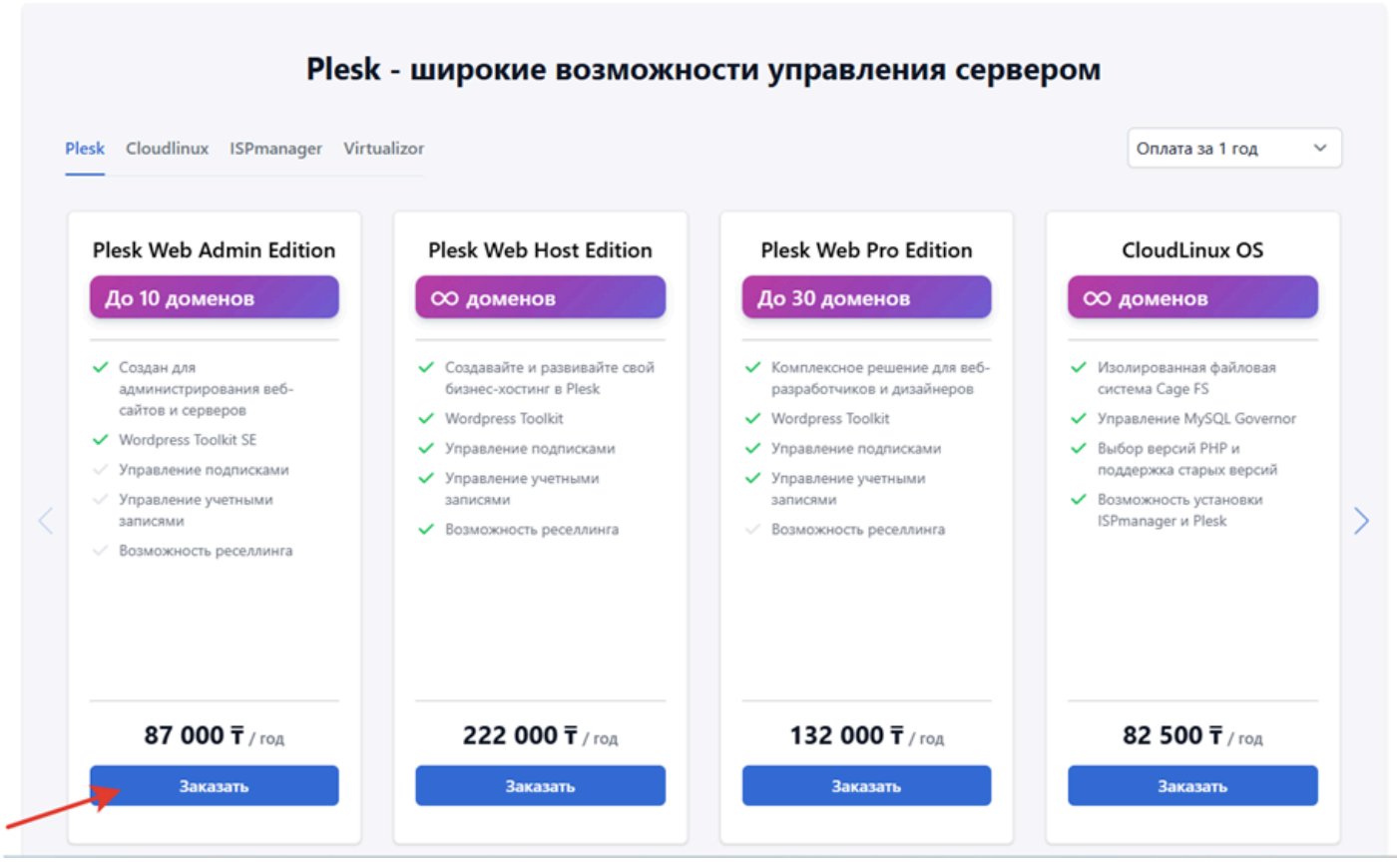
Component	Technology	Justification
Client platform	Web (mobile-first, responsive)	Flexible; we do not need separate Android and iOS apps.
Web languages and frameworks	HTML, CSS, JavaScript, React	Much cheaper and easier to build the UI on the web with React.
Server-side technology	Python, FastAPI, Uvicorn, Nginx, Playwright / BeautifulSoup	Fast to write; Playwright / BeautifulSoup access LMS data if no official API is given.
Database	None persistent (PostgreSQL only if it becomes needed)	The project only re-designs MOODLE and myDU; PostgreSQL is stable if a database is needed.
Hosting / cloud platform	GitHub Pages (UI), ps.kz (server)	GitHub Pages is free, easy and stable; ps.kz keeps data in Kazakhstan.
Development tools	Git / GitHub, VS Code, draw.io, Figma	Free tools for teamwork, diagrams and wireframes.

C3. Cloud service comparison

Criterion	ps.kz	AWS	Google Cloud
Free quota	No permanent free tier; paid hosting / VPS plans	Limited free tier or credits for new accounts	Trial credits and a few always-free products
Data location	Data centres in Kazakhstan	No data centre in Kazakhstan	No data centre in Kazakhstan
Collaboration and sharing	Team access to the control panel and SSH	IAM users and roles	IAM roles, Google account sharing
Synchronisation	Deploy from Git over SSH	Git-based deployment tools	Git-based deployment tools
Platform support	Linux VPS: Python, Nginx, any stack	All major stacks	All major stacks
Security features	SSH keys, firewall, HTTPS certificate	Two-factor authentication, encryption at rest and in transit	Two-step verification, encryption at rest and in transit
Cost after free tier	Fixed monthly fee	Pay as you go	Pay as you go
Legal fit (KZ law)	Yes	No	No

Chosen service: ps.kz.

We choose ps.kz. It is the only option that keeps the data processing inside Kazakhstan, as the law requires, and it is simple to manage by SSH. AWS is excluded because it has no data centre in Kazakhstan. Google Cloud is excluded for the same reason and because its response was too slow in our tests. When the server resources of the chosen plan are exceeded, we upgrade the plan. GitHub Pages is used only to publish the static UI files; they contain no user data.



C4. Data flow

Scenario 1 *(the user is already logged in, the official LMS API is available):*

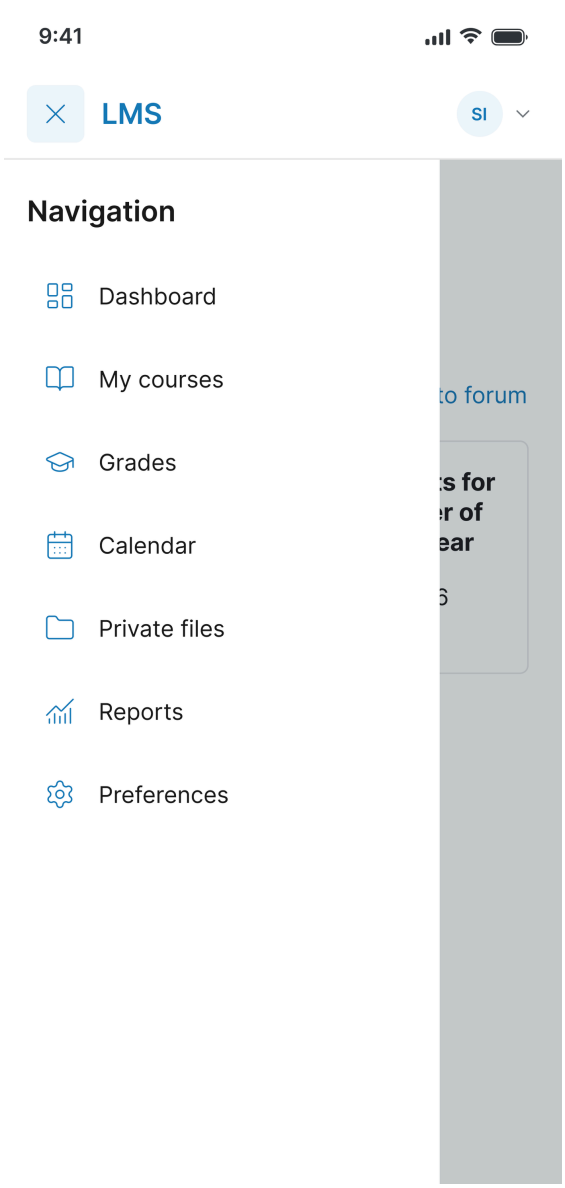
- The user taps the "Dashboard" button
- The phone sends the request over Wi-Fi or the mobile network, and DNS resolves our domain name to the IP address of the ps.kz server.
- The browser opens an HTTPS (TLS) connection and sends the request with a session token.
- Nginx passes it to FastAPI, which checks the token and calls the LMS API on behalf of the user.
- The LMS queries its own database and returns the courses and deadlines.
- Our server converts the data to JSON and returns it in the HTTPS response, and React renders the dashboard.

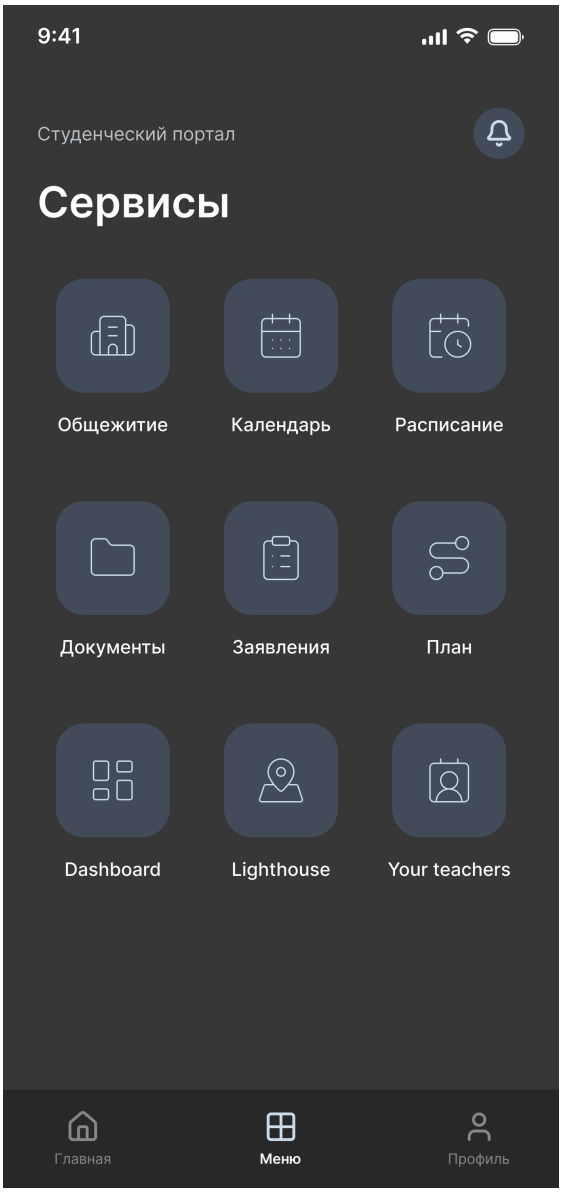
Storage: we have no persistent database; the data exists only in server memory during the session and in the university's LMS database.

Protection: in transit by TLS encryption; at rest the server disk will be encrypted and holds no user data.

Scenario 2 *(no API):* Playwright logs in to the LMS and parses the page => all other steps are the same.

Part D – Interface concept





The proposed interface helps both user groups achieve their goals by reducing the fragmentation between university platforms. **Students** can use the central dashboard to access important services such as the calendar, schedule, documents, Lighthouse and their courses without repeatedly searching through different platforms. The LMS navigation menu also provides quick access to courses, grades, calendar and other learning resources. **Administrators** benefit from a more organised and centralised interface that can provide students with consistent access to integrated university services and reduce usability complaints. Overall, the design addresses the main problem identified in the survey: students need a simpler way to find academic information across multiple platforms.

Part E – Plan for Stage 2

E1. Work plan

Week	Actions	By whom	Result
6	consult with the administration	Azhar	access to the database
7	set up local or cloud machine	Nurassyl	foundation of the application
8	design, UX/UI	Salima	layout of the platform
9	touch ups of the product	whole team	presentation of the prototype

E2. Risks

Risks	Likelihood	Impact	How to reduce it
no access to the database	67%	either demo accounts or unoptimised application	talk to the administration again and try to convince them
local machines down	60%	nothing is gonna work and we will get 0	find a place to locate the local machines or buy a cloud server
running out of time	50/50	we will get less points	time management